

Derivation of the Analytic Bikerman Counterion Closure

PNPInverse / Forward solver

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Abstract

We derive the analytic concentration profile used in `Forward/bv_solver/boltzmann.py` for inert supporting-electrolyte counterions. Starting from the electrochemical potential of an ideal dilute species we obtain the classical Boltzmann distribution $c = c_b \exp(-z\phi)$. We then replace the ideal entropy with the Bikerman lattice-gas correction $-\ln \theta$, derive the resulting algebraic closure for a single inert counterion in equilibrium with a prescribed potential $\phi(x)$ and a set of dynamic species $\{c_i(x)\}$, and finally generalize to several inert counterions sharing a common packing fraction. The result is the steady-state algebraic reduction of the dynamic Bikerman–Poisson–Nernst–Planck (BPNP) problem in which the counterion equation is eliminated exactly.

1 Setup and notation

Nondimensionalization. All quantities below are nondimensional with the same scales used in the solver. Potentials are scaled by the thermal voltage $V_T = k_B T / e$, concentrations by a reference c_{ref} , and steric coefficients by $1/c_{\text{ref}}$. Dropping tildes, we write

$$\phi \leftarrow \frac{e \phi^{\text{dim}}}{k_B T}, \quad c \leftarrow \frac{c^{\text{dim}}}{c_{\text{ref}}}, \quad a \leftarrow a^{\text{dim}} c_{\text{ref}}. \quad (1)$$

The Boltzmann factor for an ion of charge z in potential ϕ is $\exp(-z\phi)$ (no explicit $k_B T$). The product $a c$ is the local volume fraction occupied by that species.

Species partition. We split the chemical species into

$$\mathcal{D} = \{i = 1, \dots, N_d\} \quad \text{dynamic species (each has its own NP equation),} \quad (2)$$

$$\mathcal{A} = \{k = 1, \dots, N_a\} \quad \text{inert analytic counterions (no electrode reaction, no flux).} \quad (3)$$

In the production stack $\mathcal{D} = \{\text{O}_2, \text{H}_2\text{O}_2, \text{H}^+\}$ and $\mathcal{A} \subseteq \{\text{K}^+, \text{SO}_4^{2-}, \text{Cs}^+, \text{OH}^-\}$. For each $k \in \mathcal{A}$ the bulk concentration $c_{b,k}$, charge z_k , and steric coefficient a_k are fixed inputs; the spatially varying $\phi(x)$ is treated as given by the outer Poisson solve.

Bulk packing fractions. Let

$$A_{\text{dyn}}(x) = \sum_{i \in \mathcal{D}} a_i c_i(x), \quad A_{\text{dyn}}^{\text{bulk}} = \sum_{i \in \mathcal{D}} a_i c_{b,i}, \quad (4)$$

$$A_{\text{an}}^{\text{bulk}} = \sum_{k \in \mathcal{A}} a_k c_{b,k}, \quad \theta_b = 1 - A_{\text{dyn}}^{\text{bulk}} - A_{\text{an}}^{\text{bulk}}. \quad (5)$$

The bulk free-volume fraction θ_b is a constant determined entirely by configuration. The closure below is well posed only if $\theta_b > 0$, which is enforced at config-parse time.

2 Ideal Boltzmann distribution

For a dilute species in a potential ϕ the electrochemical potential (in nondimensional form) is [3]

$$\mu_i^{\text{id}}(x) = \ln c_i(x) + z_i \phi(x). \quad (6)$$

The standard-state contribution $\mu_i^{(0)}$ is suppressed because it drops out of every equation we use.

Equilibrium of an inert species. An “inert” species k has

1. no source or sink anywhere in the domain,
2. zero normal flux on every non-bulk boundary,
3. Dirichlet $c_k = c_{b,k}$ on the bulk boundary, where $\phi = 0$.

Its Nernst–Planck flux is $\mathbf{J}_k = -D_k c_k \nabla \mu_k$. At steady state the divergence vanishes and the boundary conditions above force $\mathbf{J}_k \equiv \mathbf{0}$. With $c_k > 0$ this gives

$$\nabla \mu_k(x) = 0 \quad \implies \quad \mu_k(x) = \mu_k^{\text{bulk}}. \quad (7)$$

Boltzmann profile. Substituting (6) into (7),

$$\ln c_k(x) + z_k \phi(x) = \ln c_{b,k} + z_k \cdot 0, \quad (8)$$

so

$$\boxed{c_k(x) = c_{b,k} \exp(-z_k \phi(x))}. \quad (9)$$

This is the legacy `steric_mode="ideal"` closure. It is unbounded: for an anion ($z_k = -1$) at large anodic ϕ , c_k grows exponentially, eventually exceeding any physical packing limit.

3 Bikerman chemical potential

The ideal entropy $\ln c_i$ in (6) assumes ions are point particles. Bikerman’s lattice-gas correction accounts for the finite volume each ion occupies by treating the solvent + ions as a fully packed lattice with site fraction a_i per ion. Minimizing the resulting free energy [2] gives the modified chemical potential

$$\mu_i(x) = \ln c_i(x) + z_i \phi(x) - \ln \theta(x), \quad (10)$$

where

$$\theta(x) = 1 - \sum_{j \in \mathcal{D} \cup \mathcal{A}} a_j c_j(x) \quad (11)$$

is the local free-volume fraction. The extra term $-\ln \theta$ is the excluded-volume contribution to the free energy: as ions pack into a region $\theta \rightarrow 0$, $\mu_i \rightarrow +\infty$, and the species is electrostatically expelled.

Sign convention. The codebase uses the *sign-corrected* form $\mu_{\text{steric}} = -\ln \theta$ consistently in `forms_logc.py` (`forms_logc.py:402`). The counterion closure below assumes the same sign; mixing the legacy $+\ln \theta$ branch (still present in `Forward/dirichlet_solver.py` and `robin_solver.py`, neither of which is in the production dispatch) with the closure here would be inconsistent.

4 Single-counterion algebraic closure

Consider one inert counterion ($N_a = 1$, label it “ $-$ ”). Equation (7) together with (10) gives

$$\ln c_-(x) + z_- \phi(x) - \ln \theta(x) = \ln c_{b,-} - \ln \theta_b, \quad (12)$$

i.e.

$$\frac{c_-(x) \exp(z_- \phi(x))}{\theta(x)} = \frac{c_{b,-}}{\theta_b}. \quad (13)$$

Introduce the convenient abbreviations

$$q(x) = \exp(-z_- \phi(x)), \quad B(x) = \frac{c_{b,-} q(x)}{\theta_b}. \quad (14)$$

Then (13) reads $c_- = B \theta$. Splitting $\theta = 1 - A_{\text{dyn}} - a_- c_-$ and isolating c_- :

$$c_- = B(1 - A_{\text{dyn}} - a_- c_-), \quad (15)$$

$$c_- (1 + a_- B) = B(1 - A_{\text{dyn}}), \quad (16)$$

$$\boxed{c_-(x) = \frac{c_{b,-} q(x) (1 - A_{\text{dyn}}(x))}{\theta_b + a_- c_{b,-} q(x)}}. \quad (17)$$

This is the formula implemented in `build_steric_boltzmann_expressions` (Forward/bv_solver/boltzmann.py:2

Limits. Three checks recover the expected physics.

1. *Bulk recovery.* At $\phi = 0$, $q = 1$, and $A_{\text{dyn}} = A_{\text{dyn}}^{\text{bulk}}$, so

$$c_-^{\text{bulk}} = \frac{c_{b,-} (1 - A_{\text{dyn}}^{\text{bulk}})}{\theta_b + a_- c_{b,-}} = c_{b,-}, \quad (18)$$

since $\theta_b + a_- c_{b,-} = 1 - A_{\text{dyn}}^{\text{bulk}}$ ((5) with $N_a = 1$).

2. *Dilute limit.* If $a_- \rightarrow 0$ and $A_{\text{dyn}} \rightarrow 0$ then $\theta_b \rightarrow 1$ and (17) collapses to $c_- = c_{b,-} q = c_{b,-} \exp(-z_- \phi)$, the ideal Boltzmann profile (9).
3. *High-field saturation.* For an anion at large anodic ϕ , $q \rightarrow \infty$ and the denominator is dominated by $a_- c_{b,-} q$, giving

$$c_-(x) \rightarrow \frac{1 - A_{\text{dyn}}(x)}{a_-}. \quad (19)$$

The counterion fills exactly the free volume left by the dynamic species, never more. Contrast with the unbounded ideal Boltzmann, where $c_- \sim c_{b,-} q$ diverges.

Total packing. Substituting (17) into $\theta = 1 - A_{\text{dyn}} - a_- c_-$ yields

$$\theta(x) = (1 - A_{\text{dyn}}(x)) \frac{\theta_b}{\theta_b + a_- c_{b,-} q(x)}, \quad (20)$$

which is strictly positive whenever $1 - A_{\text{dyn}} > 0$ and $\theta_b > 0$.

5 Multi-counterion shared- θ closure

When $|\mathcal{A}| > 1$ each analytic counterion satisfies its own zero-flux equilibrium, but all of them share the same packing fraction θ . Writing (13) for each $k \in \mathcal{A}$ and abbreviating

$$q_k(x) = \exp(-z_k \phi(x)), \quad B_k(x) = \frac{c_{b,k} q_k(x)}{\theta_b}, \quad (21)$$

we have one algebraic equation per counterion,

$$c_k = B_k \theta = B_k \left(1 - A_{\text{dyn}} - \sum_{k' \in \mathcal{A}} a_{k'} c_{k'} \right). \quad (22)$$

Reduction via the analytic packing sum. Let $S(x) = \sum_{k \in \mathcal{A}} a_k c_k(x)$. Multiplying (22) by a_k and summing,

$$S = \left(\sum_k a_k B_k \right) (1 - A_{\text{dyn}} - S), \quad (23)$$

hence

$$S = (1 - A_{\text{dyn}}) \frac{\sum_k a_k B_k}{1 + \sum_k a_k B_k}. \quad (24)$$

The remaining free volume is

$$1 - A_{\text{dyn}} - S = \frac{1 - A_{\text{dyn}}}{1 + \sum_k a_k B_k} = \frac{\theta_b (1 - A_{\text{dyn}})}{\theta_b + \sum_k a_k c_{b,k} q_k(x)}. \quad (25)$$

Substituting back into (22) gives the closed-form expression for each counterion:

$$\boxed{c_k(x) = \frac{c_{b,k} q_k(x) (1 - A_{\text{dyn}}(x))}{\theta_b + \sum_{k' \in \mathcal{A}} a_{k'} c_{b,k'} q_{k'}(x)}, \quad k \in \mathcal{A}.} \quad (26)$$

With $A_{\text{dyn}} \equiv 0$ this collapses to the standard full-equilibrium asymmetric-Bikerman ion-distribution formula of [2]; the $(1 - A_{\text{dyn}})$ pre-factor is the contribution of the dynamic species' local packing, retained because they obey their own NP equations rather than being in equilibrium with ϕ . The relation is worked out in detail in §6.

Structure of the closure.

- The numerator $c_{b,k} q_k (1 - A_{\text{dyn}})$ is *per-ion*: each counterion has its own Boltzmann factor and the common dynamic-species free-volume multiplier.
- The denominator $\theta_b + \sum_{k'} a_{k'} c_{b,k'} q_{k'}$ is *shared* across all k . This is exactly what makes the multi-ion problem decouple: the algebra never requires solving a coupled local nonlinear system.
- For $|\mathcal{A}| = 1$ the denominator reduces to $\theta_b + a_- c_{b,-} q$ and (26) collapses to (17).
- For two oppositely charged ions (e.g. K^+ with $z = +1$ and SO_4^{2-} with $z = -2$) the denominator contains both $\exp(-\phi)$ and $\exp(+2\phi)$, so it is bounded away from zero on both sides of $\phi = 0$.

High-field saturation, multi-ion. For large ϕ of either sign, one term in $\sum_{k'} a_{k'} c_{b,k'} q_{k'}$ dominates. Call its index $k^* = \arg \max_{k'} (-z_{k'} \phi)$. Then

$$c_{k^*}(x) \longrightarrow \frac{1 - A_{\text{dyn}}(x)}{a_{k^*}}, \quad (27)$$

$$c_k(x) \longrightarrow 0 \quad \text{for } k \neq k^*, \quad (28)$$

i.e. the most strongly attracted analytic counterion fills all of the remaining free volume and the others are depleted. This is the multi-ion analog of (19).

6 Relation to Tresset's generalized Poisson–Fermi formula

The full-equilibrium limit ($A_{\text{dyn}} \equiv 0$) of (26) is not new: it coincides with Tresset's generalized Fermi-like ion distribution for size-asymmetric multi-species lattice gases [2, Eq. (19), page 061506-4]. The hybrid extension we use here — closing only the inert subset analytically while the reactive species remain dynamic — amounts to one additional algebraic step that introduces the explicit $(1 - A_{\text{dyn}}(x))$ pre-factor.

Tresset's full-equilibrium formula. Minimizing the lattice-gas free energy for M species of valence z_i and excluded volume v_i and imposing thermodynamic equilibrium $\mu_i(\mathbf{r}) = \mu_i^{\text{bulk}}$ for *every* species, Tresset [2, Eq. (19)] obtains

$$\phi_i(x) = \frac{\exp(-z_i \psi^*(x)) \phi_i^\infty}{\phi_{\text{FS}}^\infty + \sum_{j=1}^M \exp(-z_j \psi^*(x)) \phi_j^\infty}, \quad 1 \leq i \leq M, \quad (29)$$

where $\phi_i = c_i v_i$ is the local volume fraction of species i , $\phi_i^\infty = c_i^\infty v_i$ its bulk value, $\phi_{\text{FS}}^\infty = 1 - \sum_j \phi_j^\infty$ the bulk free-space volume fraction, and $\psi^* = e\psi/k_B T$ the dimensionless electrostatic potential.

Notation map.

Tresset [2]	This work
v_i (excluded volume)	a_i (nondim steric coefficient)
$\phi_i = c_i v_i$	$a_i c_i$
$\phi_i^\infty = c_i^\infty v_i$	$a_i c_{b,i}$
$\phi_{\text{FS}}^\infty = 1 - \sum_j \phi_j^\infty$	θ_b (5)
$\psi^*(x) = e\psi(x)/k_B T$	$\phi(x)$ (already nondim)
$\exp(-z_i \psi^*)$	$q_i = \exp(-z_i \phi)$

Notation caveat: Tresset uses ϕ for volume fraction and ψ for potential; we use ϕ for potential. The collision is unfortunate but the dictionary above is unambiguous.

Equivalence on the full-equilibrium subset. Set $A_{\text{dyn}} \equiv 0$ in (26), multiply both sides by a_k , and apply the dictionary:

$$a_k c_k(x) = \frac{a_k c_{b,k} q_k(x)}{\theta_b + \sum_{k'} a_{k'} c_{b,k'} q_{k'}(x)} \xleftrightarrow{\text{map}} \phi_k(x) = \frac{\phi_k^\infty e^{-z_k \psi^*}}{\phi_{\text{FS}}^\infty + \sum_{k'} \phi_{k'}^\infty e^{-z_{k'} \psi^*}}, \quad (30)$$

which is (29) verbatim. The algebraic steps in Tresset (his Eqs. (15)–(19), pages 061506-3 to 061506-4) coincide with §5 of this note: identical chemical potential $\mu_i = \ln c_i + z_i \phi - \ln \theta$, identical equilibrium condition $\nabla \mu_i = 0$, identical algebraic inversion.

Modification for hybrid coupling. Tresset’s (29) assumes *every* species is in equilibrium with ϕ . In the modified-PNP problem the dynamic species $i \in \mathcal{D}$ are not: they obey Nernst–Planck equations with electrode-reaction source terms, and their profiles $c_i(x)$ are determined by transport and kinetics. The zero-flux argument $\nabla\mu_i = 0$ (7) *fails for them*; it still holds for the inert subset $k \in \mathcal{A}$.

Repeating Tresset’s algebra on the inert subset only, with the dynamic species’ packing $A_{\text{dyn}}(x) = \sum_{i \in \mathcal{D}} a_i c_i(x)$ treated as a known spatial background, the equilibrium condition for $k \in \mathcal{A}$ becomes

$$\ln c_k(x) + z_k \phi(x) - \ln[1 - A_{\text{dyn}}(x) - S(x)] = \ln c_{b,k} - \ln \theta_b, \quad (31)$$

with $S(x) = \sum_{k \in \mathcal{A}} a_k c_k(x)$ the analytic-counterion packing. The only difference from Tresset’s intermediate step is that the local free-space fraction $\phi_{\text{FS}}(x) = 1 - \sum_j \phi_j(x)$ has been split into a dynamic part $A_{\text{dyn}}(x)$ (now a spatial background, not a sum-of-equilibrium-profiles) and an inert part $S(x)$ (the unknown). Following exactly the same multiply-by- a_k /sum/isolate procedure as in §5 returns

$$c_k(x) = \frac{c_{b,k} q_k(x) (1 - A_{\text{dyn}}(x))}{\theta_b + \sum_{k' \in \mathcal{A}} a_{k'} c_{b,k'} q_{k'}(x)}, \quad k \in \mathcal{A}, \quad (32)$$

which is (26). The lone structural difference from Tresset’s (29) is the explicit $(1 - A_{\text{dyn}}(x))$ factor in the numerator.

When the modification matters. If the dynamic species’ local packing equals its bulk value everywhere ($A_{\text{dyn}}(x) \equiv A_{\text{dyn}}^{\text{bulk}}$), the pre-factor is a global constant. Defining $\theta'_b = \theta_b/(1 - A_{\text{dyn}}^{\text{bulk}})$ and $c'_{b,k} = c_{b,k}/(1 - A_{\text{dyn}}^{\text{bulk}})$ absorbs it into rescaled bulk quantities and (32) reduces to Tresset’s (29) exactly. The correction matters precisely when $A_{\text{dyn}}(x)$ varies in space, which is the rule, not the exception, in PNP problems with electrode reactions. In the production stack the proton H^+ Boltzmann-piles up at the OHP under cathodic polarization, so $A_{\text{dyn}}(\text{OHP}) \gg A_{\text{dyn}}^{\text{bulk}}$ and the inert counterion is volumetrically squeezed out of the diffuse layer.

What is and is not novel. Equation (29) — the size-asymmetric, all-equilibrium lattice-gas distribution — is Tresset [2]. The hybrid-coupling extension (32) = (26) — closing the inert subset analytically while keeping the reactive species dynamic, with the explicit $(1 - A_{\text{dyn}}(x))$ pre-factor — is, to our knowledge, not previously written in closed form in the modified-PNP literature. The algebraic modification is one step; the use case (reactive analyte species coupled to an inert supporting electrolyte through a closed-form analytic counterion in a steric-aware modified-PNP problem) appears to be original to this work.

7 Poisson coupling and remaining BVP

The closure (26) eliminates the analytic counterion(s) from the dynamic problem. What remains is

(i) **Poisson with closed-form counterion charge.**

$$-\nabla \cdot (\varepsilon \nabla \phi) = \rho_{\text{chg}} \left(\sum_{i \in \mathcal{D}} z_i c_i + \sum_{k \in \mathcal{A}} z_k c_k[\phi, \{c_i\}] \right), \quad (33)$$

where ρ_{chg} is the nondimensional charge prefactor (`charge_rhs_prefactor`) and the $c_k[\phi, \{c_i\}]$ are the explicit functionals (26).

(ii) **Nernst–Planck for dynamic species with full packing.** For each $i \in \mathcal{D}$, the modified Poisson–Nernst–Planck flux [1] reads

$$\partial_t c_i + \nabla \cdot \mathbf{J}_i = R_i, \quad (34)$$

$$\mathbf{J}_i = -D_i c_i \nabla \mu_i, \quad \mu_i = \ln c_i + z_i \phi - \ln \theta, \quad (35)$$

with the *total* packing

$$\theta(x) = 1 - A_{\text{dyn}}(x) - \sum_{k \in \mathcal{A}} a_k c_k [\phi, \{c_i\}], \quad (36)$$

which by (25) simplifies to

$$\theta(x) = \frac{\theta_b (1 - A_{\text{dyn}}(x))}{\theta_b + \sum_{k \in \mathcal{A}} a_k c_{b,k} q_k(x)}. \quad (37)$$

This is the key cross-check on consistency: *the same* closure expression must appear in (a) the Poisson charge density and (b) the dynamic-species steric chemical potential. Wiring them from distinct expressions would amount to solving two different models, and was the failure mode that prompted the refactor of the original ideal-Boltzmann path (see `docs/solver/steric_analytic_clo4_reduction_handoff.md`).

8 Equivalence to dynamic Bikerman–NP

The reduction (26) is *not* an approximation: it is the exact steady-state solution of the dynamic Bikerman–NP equation for each inert counterion, provided

1. the system is at steady state,
2. each counterion has no electrode reaction, no source/sink, and no imposed flux on any non-bulk boundary,
3. the bulk boundary fixes $c_k = c_{b,k}$ and $\phi = 0$,
4. the same Bikerman chemical potential (10) acts on dynamic and counterion species, and
5. the total packing θ in the dynamic species’ residual includes the analytic counterion’s contribution.

Under these assumptions an explicit calculation shows $\mathbf{J}_k = 0$ for the closure profile, so it is the unique solution of the counterion’s Nernst–Planck equation up to a multiplicative gauge that the bulk Dirichlet condition fixes.

The reduction is *not* equivalent in any of the following situations and should not be used: (a) transient double-layer charging with finite analyte flux through the supporting electrolyte; (b) ion-specific adsorption or chemisorption on the electrode; (c) models with a finite closest-approach plane treated dynamically rather than as a Stern boundary condition.

9 Numerical implementation notes

Three numerical guards in the implementation are worth noting because they appear in the production UFL but are not part of the formal derivation.

Potential clamp. The Boltzmann factor $q_k = \exp(-z_k \phi)$ is replaced with $\exp(-z_k \tilde{\phi})$ where $\tilde{\phi} = \min(\max(\phi, -\phi_{\text{clamp},k}), \phi_{\text{clamp},k})$. This is a Newton-stability guard during continuation; on a converged solution $\tilde{\phi} \equiv \phi$ provided $|\phi|$ stays inside the clamp. The clamp is per-ion to allow asymmetric bounds. The shared denominator in (26) uses each ion's own clamp for its own term, which is consistent because each $\phi_{\text{clamp},k}$ only ever appears multiplied by z_k inside the k -th summand.

Free-volume floor. The factor $1 - A_{\text{dyn}}$ in the numerator is replaced by $\max(1 - A_{\text{dyn}}, \epsilon_{\text{floor}})$ with $\epsilon_{\text{floor}} = 10^{-10}$. This is a finite-element interpolation safety: even with $\theta > 0$ pointwise in the analytic solution, the FE basis can produce small negative $(1 - A_{\text{dyn}})$ values on coarse meshes inside Newton iterates. The floor is never active on a converged, physically meaningful solution.

z -ramp scaling. The Poisson charge contribution $z_k c_k[\phi, \{c_i\}]$ is multiplied by a shared scaling Function $z_{\text{scale}} \in [0, 1]$. Continuation strategies that ramp dynamic-species charges from 0 to z_i also ramp the analytic counterion's charge through this factor, keeping the bulk electroneutral throughout the homotopy. Default $z_{\text{scale}} = 1$ recovers the physical model.

10 Code map

Mathematical object	Location
$\mu_i = \ln c_i + z_i \phi - \ln \theta$	Forward/bv_solver/forms_logc.py:402
c_k (26)	boltzmann.py:254 (c_steric_expr)
$a_k c_k$ packing contribution	boltzmann.py:257 (packing_contribution)
$z_k c_k$ Poisson source	boltzmann.py:258 (charge_density)
$\theta_b > 0$ validation	boltzmann.py:159--171
Double-counting validation	boltzmann.py:176--201
Ideal closure $c_k = c_{b,k} \exp(-z_k \phi)$	boltzmann.py:380--382

Configuration entry. Each analytic counterion is declared as a dict in `bv_bc.boltzmann_counterions`, e.g.

```
{
  "z": +1,
  "c_bulk_nondim": C_KPLUS_HAT,
  "a_nondim":      A_KPLUS_HAT,
  "phi_clamp":     50.0,
  "steric_mode":   "bikerman",
}
```

with `steric_mode="ideal"` selecting the legacy closure (9) and `steric_mode="bikerman"` selecting (26).

References

- [1] M. S. Kilic, M. Z. Bazant, and A. Ajdari. Steric effects in the dynamics of electrolytes at large applied voltages. II. Modified Poisson–Nernst–Planck equations. *Physical Review E*, 75(2):021503, 2007. <https://doi.org/10.1103/PhysRevE.75.021503> arXiv preprint: <https://arxiv.org/abs/physics/0611232>

- [2] G. Tresset. Generalized Poisson–Fermi formalism for investigating size correlation effects with multiple ions. *Physical Review E*, 78(6):061506, 2008. <https://doi.org/10.1103/PhysRevE.78.061506>
- [3] J. Newman and K. E. Thomas-Alyea. *Electrochemical Systems*. Wiley-Interscience, Hoboken NJ, 3rd edition, 2004.